

**Q 1.** List the five layers of the Internet protocol stack. For each layer, offer a concise explanation using one sentence, and give an example of a protocol or specific networking technology of that layer.

① Application Layer :

Provides network services directly to user applications, 为用户应用程序直接提供网络服务,

enabling communication like email, web browsing, and file transfer. 使得像电子邮件、网页浏览、文件传输等通信成为可能。

→ HTTP, SMTP, DNS

② Transport Layer :

Ensures reliable or efficient end-to-end data delivery between processes on different hosts.

在不同主机的进程之间提供端到端的数据传输, 保证数据传送的可靠性, 高效性。

→ TCP, UDP

③ Network Layer :

Determines the path that data takes through the network and handles logical addressing.

决定数据在网络中传输的路径, 并负责逻辑地址(IP 地址)的管理

→ IP (Internet Protocol)

④ Link Layer :

Handles data transfer between directly connected devices and manage physical addressing.

负责在直接相连的设备之间进行数据帧的传输, 并处理物理地址(Mac 地址)

→ Ethernet, PPP

⑤ Physical Layer :

Transmits raw bits over a physical medium, such as electrical / optical signals.

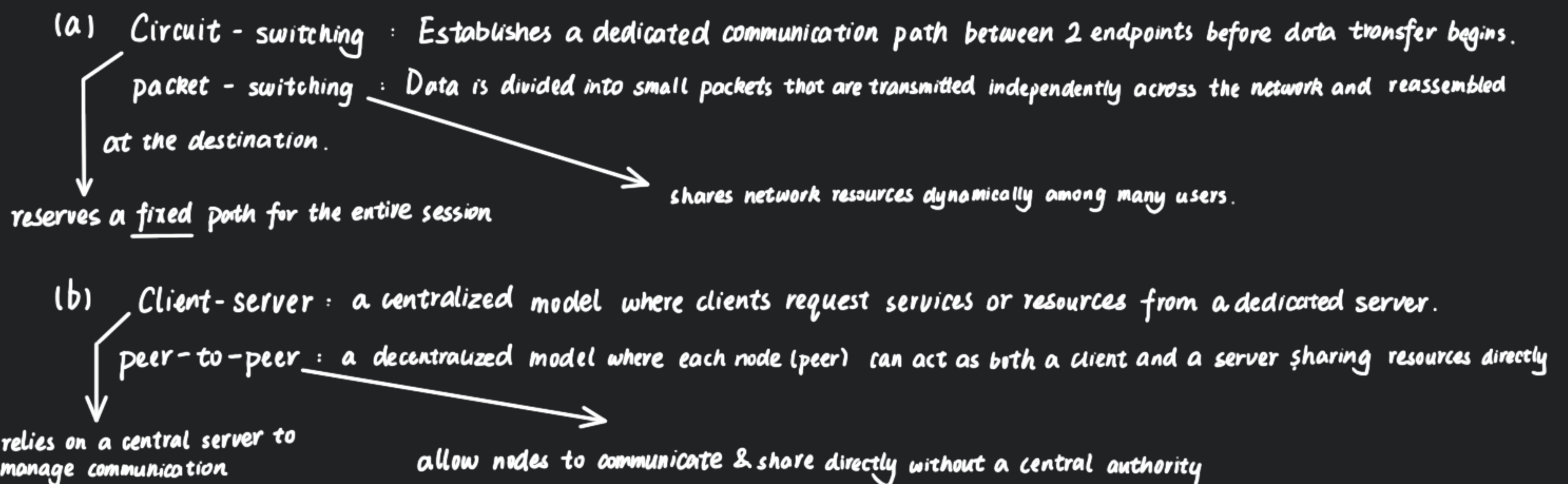
负责在物理介质上传输比特流, 例如电信号/光信号

→ Wi-Fi, Fiber optic cable, Ethernet cable

**Q 2.** For the following pairs of technical terms, provide a definition for each and explain the primary differences between them. Your response should be clear and concise.

(a) "Circuit-switching" and "packet-switching"

(b) "Client-server" and "peer-to-peer"





Q 3. Answer the following questions:

- What type of applications is TCP better suited for? Any application examples?
- What type of applications is UDP better suited for? Any application examples?

(a) TCP is better suited for applications that require reliable data transmission, error checking, and ordered delivery of packets.

→ Web browsing (HTTP/HTTPS), email (SMTP), file transfer (FTP)

(b) UDP is better suited for applications that need fast implementation and can tolerate some data loss, without the overhead of reliability mechanisms.

→ video streaming, online gaming, voice over IP

Q 4. Consider a packet of  $L$  bits sending from the source to destination through a  $K$ -hop path. That is, there are  $K - 1$  routers between the source and destination. Suppose each link has a transmission rate of  $R$  bits per



second (bps), and the propagation delay is  $d$  for each hop.

- Consider a packet switching network. Suppose there is no nodal processing delay and queuing delay. What is the end-to-end delay?

$$t_1 = K \cdot \left( \frac{L}{R} + d \right)$$

端对端延迟
传播延迟
发送延迟

- Consider a circuit switching network. Suppose the circuit setup time is  $\tau$  seconds. Links in the network use frequency division multiplexing (FDM), where the associated frequency band is divided into  $F$  subbands, each being allocated to a user (or circuit). [Hint: As a result, the transmission rate allocated to each user (or circuit) is  $R/F$ .] What is the end-to-end delay?

$$t_2 = \tau + \frac{LF}{R} + Kd$$

- Consider a packet switching network with  $L = 1000$  bits,  $K = 2$ ,  $R = 20$  Mbps,  $d = 10\mu s$ . There are two packets sent one after the other, and there are no other packet in the system. Let the nodal processing delay at the router be  $5\mu s$ . We ignore the nodal processing delay at the source and destination. Compute the time required to send both packets from the source and destination. [Note: In our lecture, we set 1 Kbit =  $10^3$  bits and 1 Mbit =  $10^6$  bits.]

与 packet-switching 不同, 在 circuit-switching 中, 每 1 位 bit 从源端发出后, 可直接以传播速度通过所有中间节点 (无需每一跳重新发送/排队), 因为电路已提前建好并被独占。

$$(c) \text{ First packet: } \frac{1000}{20 \times 10^6} s + 10\mu s + 5\mu s + \frac{1000}{20 \times 10^6} s + 10\mu s = 125\mu s$$

The second packet starts its first transmission right after the first packet finishes its first transmission (at  $50\mu s$ ).

$$t_3 = 50\mu s + 125\mu s = 175\mu s$$

Q 5. Consider a set of packets with size 10 Mbits and a queue. These packets arrive at the queue with certain patterns defined in (a) and (b), waiting for transmission. The transmission rate is 10 Mbps.



- Suppose there is one packet arrival every second. What is the average queuing delay of these packets?

- Suppose  $K$  packets arrive simultaneously every  $K$  seconds. What is the average queuing delay of these packets?

- What are the traffic intensity of the scenarios considered in (a) and (b)? Any insights?

(a) each packet takes  $\frac{10 \text{ Mbits}}{10 \text{ Mbps}} = 1s$ , packets arrive every 1 second  $\Rightarrow$  queuing delay = 0s

(b) package number: 1  $\rightarrow$  queuing delay: 0s

2  $\rightarrow$

1s

$$\Rightarrow \text{average queuing delay} = \frac{0+1+\dots+(K-1)}{K} = \frac{K \cdot (K-1)}{2K} = \frac{K-1}{2} s$$

⋮

⋮

K  $\rightarrow$

(K-1)s

(c) traffic intensity  $\rho$  = average arrival rate  $\times$  service time per ticket

$$(a) \quad 1 \text{ package/s} \times \frac{10 \text{ Mbits}}{10 \text{ Mbps}} = 1 \Rightarrow \text{even with the same average load } (\rho), \text{ bursty traffic patterns}$$

$$(b) \quad K/K = 1 \text{ package/s} \times 1 = 1 \Rightarrow \text{lead to higher queuing delays than smooth, regular arrivals.}$$



Q 6. Consider the following message and answer questions.

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host:
gaia.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (Windows;U; Windows NT
5.1; en-US; rv:1.7.2) Gecko/20040804 Netscape/7.2 (ax)
<cr><lf>Accept:ext/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5<cr><lf>Accept-Language:
en-us, en;q=0.5<cr><lf>Accept-Encoding: zip,deflate<cr><lf>Accept-
Charset: ISO-8859-1, utf-8;q=0.7,*/*;q=0.7<cr><lf>Keep-Alive:
300<cr><lf>Connection:keep-alive<cr><lf><cr><lf>
```

- Does HTTP message run on top of TCP or UDP? Why is TCP or UDP a better choice? Please explain the reason by considering the features of TCP or UDP.
- Is this message an HTTP request message or an HTTP response message?
- Does this message corresponds to a non-persistent or a persistent connection?
- Suppose the browser who sent this message was assigned an identification number 1150 by the associated host. What is the corresponding entry in the cookie file of the browser? To specific its cookie, which header line (including header field name and value) should be included in the message?
- If the server receives this message successfully and is going to return the requested object in a message, what would be the status line? What would be included in the entity body?

(a) HTTP runs on top of TCP

TCP is a better choice, because it provides reliable, connection-oriented communication, guarantees the order of data, error checking and retransmission of lost packets. HTTP requires reliable delivery of web pages and resources, so TCP's features are necessary. UDP doesn't guarantee delivery/order and isn't suitable for HTTP.

(b) request message.

∴ starts with 'GET'

(c) persistent connection

∴ contains 'Connection: keep-alive', which tells the server to keep the TCP connection open for further requests and responses.

(d) The corresponding entry in the browser's cookie file would be:

Host: gaia.cs.umass.edu

Cookie: id=1150

The header line to include in the HTTP message should be:

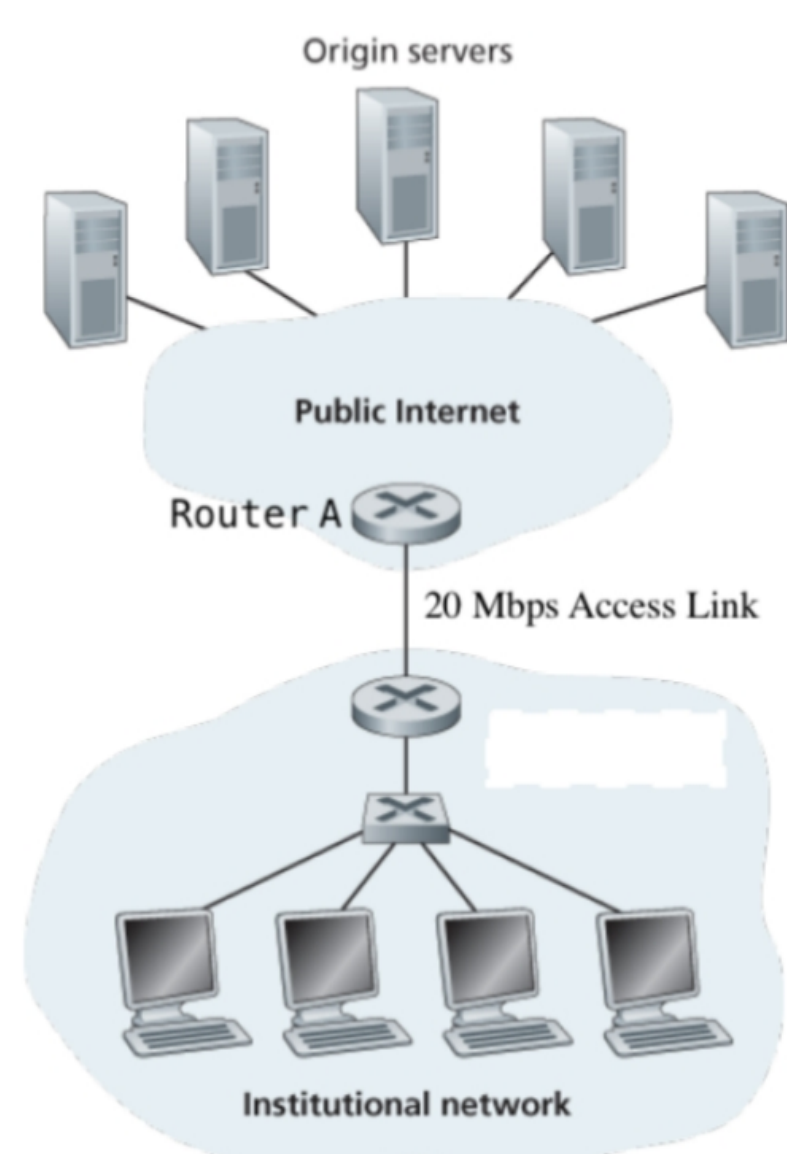
Cookie: id=1150

(e) The status line would be:

HTTP/1.1 200 OK

The entity body would contain the contents of '/cs453/index.html', i.e., the HTML file requested.

Q 7. Consider the following figure with an institutional network connected to the Internet. There are a set of objects with size 1 Mbits. Suppose the institution's browsers has an average request rate of 10 requests per second, and all those requests are sent to the origin servers. The average response time is determined as follows:



$$\text{Average Response Time} = \text{Internet Delay} + \text{Average Access Delay.} \quad (1)$$

- Internet delay is the round trip time between router A and the origin server. It is equal to 2 seconds.



- Average access delay is the delay from Router A to the institution router. The transmission rate of the access link is 20 Mbps. The average access delay is equal to  $\frac{\Delta}{1-\Delta\beta}$ , where  $\Delta$  is the average time required to transmit an object over the access link (i.e., the transmission delay of an object);  $\beta$  is the arrival rate (in requests per second) at the access link.

- Note: The delays over other links, e.g., local area network (LAN) delay, are regarded as zero.

Answer the following questions:

- (a) Derive average response time of the system.

平均响应时间 < 1s

- Suppose there is a cache installed in the institutional LAN. Compute the hit rate  $x \in [0, 1]$  that leads to an average response time that is less than 1 second.

命中缓存-响应时间近似为0 (LAN延迟忽略)

未命中时响应时间为2.1秒

$$(a) \quad \Delta = \frac{1 \text{ Mbits}}{20 \text{ Mbps}} = 0.05 \text{ s} \quad \beta = 10 \text{ requests/second}$$

$$\text{average access delay} = \frac{0.05}{1 - 0.05 \times 10} = 0.1 \text{ s}$$

$$\text{average response time} = 2 \text{ s} + 0.1 \text{ s} = 2.1 \text{ s}$$

$$(b) \quad \text{average response time} = x \cdot 0 + (1-x) \cdot (2 + \frac{0.05}{0.05 \cdot (1-x) - 10}) < 1, \text{ that is } 2x^2 + 1.1x - 1.1 > 0$$

$$x > 0.5159646$$

$\therefore$  the cache hit rate  $x$  must be greater than 0.516 to make the average response time < 1s

Q 8. Suppose you click a web page within your Web browser, and your local DNS server has stored the related resource records. Let  $RTT_0$  denote the round trip time (RTT) between your host and your local DNS server. On the web page you visit, there are an HTTP basic file and 12 referenced objects. Let  $RTT_1$  denote the RTT between the local host and the Web server. The HTTP basic file has a size of  $L$  Mbit, and each referenced object has a size of  $L$  Mbit. The transmission rate is  $R$  Mbps. Compute how much time elapses from when you click the link until your web browser receives the objects.

- (a) Non-persistent HTTP with no parallel TCP connections?
- (b) Non-persistent HTTP with the browser configured for 4 parallel connections?
- (c) Persistent HTTP? Note that in this case, if a client knows the URLs of its requested referenced objects, the client can send requests of referenced objects back-to-back without waiting for the responses.

(a) 1. the browser needs one  $RTT_0$  to query DNS and get the server's IP address

2. takes 1  $RTT_1$  to establish the TCP connection

3. takes 1  $RTT_1$  to request & receive the HTTP base file.

4. the transmission time for the base file is  $L/R$  seconds.

5. For the 12 referenced objects, each requires a separate TCP connection ( $\because$  non-persistent, no parallel)

Each object takes 2  $RTT_1$  (1 for TCP set-up, 1 for request/response), plus  $\frac{L}{R}$  for transmission,

and these 12 are done sequentially

6. Total time =  $RTT_0 + RTT_1 + RTT_1 + \frac{L}{R} + 12 \cdot (2RTT_1 + \frac{L}{R})$

$$= RTT_0 + 26RTT_1 + 13\frac{L}{R}$$

(b) For the 12 referenced objects, 4 at a time, 3 rounds of downloads

$$\text{Total time} = RTT_0 + 2RTT_1 + \frac{L}{R} + 3 \cdot (2RTT_1 + \frac{L}{R})$$

$$= RTT_0 + 8RTT_1 + 4\frac{L}{R}$$

(c) After the base file is received, all 12 object requests are sent immediately.

Each object is transmitted one after another, but the requests can be sent together.

$$\text{Total time} = RTT_0 + RTT_1 + RTT_1 + \frac{L}{R} + RTT_1 + 12 \cdot \frac{L}{R}$$

$$= RTT_0 + 3RTT_1 + 13\frac{L}{R}$$



**Q 9.** Consider distributing a file of  $F$  bits to  $N$  peers using a client-server architecture. Assume a fluid model where the server can simultaneously transmit to multiple peers, transmitting to each peer at different rates, as long as the combined rate does not exceed  $u_s$ .

- (a) Suppose that  $u_s/N \leq d_{\min}$ . Specify a distribution scheme that has a distribution time of  $NF/u_s$ .  
 (b) Suppose that  $u_s/N \geq d_{\min}$ . Specify a distribution scheme that has a distribution time of  $F/d_{\min}$ .  
 (c) Conclude that the minimum distribution time is in general given by  $\max\{NF/u_s, F/d_{\min}\}$ .

(a) The server can split its bandwidth equally among the  $N$  peers.

It takes  $\frac{F}{\frac{u_s}{N}} = \frac{FN}{u_s}$  seconds for each peer to receive the entire file.

(b) The server sends data to each peer at their maximum download rate, but the slowest peer determines the total distribution time, which is  $\frac{F}{d_{\min}}$ .

(c) In general, the minimum distribution time is given by  $\max\{\frac{NF}{u_s}, \frac{F}{d_{\min}}\}$ .

This is because the total distribution time depends on both the server's upload bandwidth, and the slowest peer's download bandwidth.

The server cannot send data faster than its upload bandwidth.

and no peer can receive data faster than its own download bandwidth.

**Q 10.** Explain why a hierarchical DNS system (instead of a huge centralized server) is used.

- ① Administrative Decentralization: Different organizations can manage their own authoritative DNS servers (e.g. google.com's DNS records are managed by Google)  $\Rightarrow$  easier to update, maintain & secure DNS records.
- ② Load Distribution: Queries are spread across many servers, preventing any one server from being overwhelmed. The hierarchical structure helps direct queries efficiently.
- ③ Fault Tolerance and Reliability: A single centralized DNS server would be a single point of failure. If it goes down, the entire Internet's name resolution would fail. The hierarchical and distributed structure ensures that even if some servers fail, the system as a whole continues to function.
- ④ Performance: By distributing DNS servers globally and organizing them hierarchically, DNS queries can be resolved more quickly. Local DNS servers can cache recent requests, reducing latency and improving user experience.
- ⑤ Scalability: The DNS system needs to serve billions of users and manage hundreds of millions of domain names. A hierarchical design allows the workload to be distributed across many servers, making the system scalable and capable of handling massive global traffic.